Requirements Verification Artifacts

Requirement IDs 1–39 | Proposed draft for team/GTA review

SCOPE STATUS The workbook leaves Sprint No. blank for all 39 rows. This document covers the full requirements backlog;

Purpose

Turn each source requirement into a reviewable verification package: expected behavior, acceptance criteria, example inputs and expected outputs, constraints, dependencies, a proposed definition of done, and a place to attach real evidence. The example tests below are proposals only; none were executed, and no implementation results are claimed.

How to review this artifacts

- Each requirement section preserves the source row wording and adds a proposed, testable interpretation.
- Items that need a policy choice (such as currency rates, variance convention, precision, or supported equation class) call it out in the validation/open-decisions section.
- After the team agrees sprint scope, enter the approved sprint number in the google sheet and copy only those approved requirements into the Sprint 1 submission package.

Google Sheet: Project 3: Requirements Open source workbook

Requirement register

ID	Requirement	Points	Priority	Sprint No.
01	Expression parser	1	1	1
02	Arbitrary-precision calculation engine	1	2	1
03	Persistent session history and Ans recall	1	3	1
04	Unit conversion	2	4	1
05	Clean, user-friendly UI layout	2	5	1
06	2D single-variable graphing	2	6	1
07	Calculation mode switcher	1	7	1
08	Function search and category filter	1	8	1
09	Financial calculations	2	9	2
10	Calculation-step and formula breakdown	2	10	2
11	Helpful input checker	1	11	2
12	Undo and redo for expression editing	1	12	2
13	Number theory tools	3	13	2
14	Displayed precision controls	2	14	2
15	Tolerance calculator	2	15	2
16	Statistical calculations	3	16	2
17	Fraction calculator	3	17	2

ID	Requirement	Points	Priority	Sprint No.
18	Percentage calculator	2	18	2
19	Graph customization	3	19	2
20	Basic algebra equation solver	5	20	2
21	Favorite calculations	2	21	2
22	Display customization	2	22	2
23	Export calculation history	3	23	2
24	Graph comparison	3	24	2
25	Financial scenario comparison	3	25	2
26	Programmer mode	2	26	2
27	Matrix operations	2	27	2
28	Calculus tools	1	28	2
29	Complex-number mode	1	29	2
30	Find graph intercepts	2	30	Unassigned
31	Constants library	2	31	Unassigned
32	Save graphs as PNG	1	32	Unassigned
33	Notes on saved calculations	1	33	Unassigned
34	Light and dark mode	1	34	Unassigned
35	Region-dependent number formatting	1	35	Unassigned
36	Right-click shortcut menu	1	36	Unassigned
37	Thousands separator toggle	1	37	Unassigned
38	Degree/radian angle toggle	1	38	Unassigned
39	Scientific notation toggle	1	39	Unassigned

Requirement 01 — Expression parser

Proposed / hypothetical • Story points: 1 • Priority: 1 • Sprint: 1

Source description: A core expression parser handling basic arithmetic, scientific functions, and parenthetical evaluation.

What it should do

Read a calculator expression, apply the agreed grammar and operator precedence, and pass a structured expression to the calculation engine. Support arithmetic, parentheses, and the scientific functions the team selects.

Acceptance criteria

- Multiplication and division take precedence over addition and subtraction; parentheses override precedence.
- Recognized scientific functions are parsed with the agreed argument and angle-unit conventions.
- Equivalent valid expressions produce the same expression structure regardless of harmless whitespace.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R01-T1	2 + 3 × 4	14; multiplication is evaluated before addition.
R01-T2	(2 + 3) × 4	20; the parenthesized sum is evaluated first.
R01-T3	sin(π/2)	1, if radians and the listed sin function are selected; confirm angle units with the team.

UI / workflow notes: One expression entry area; preserve the entered expression while showing the parsed/evaluated result or a linked diagnostic.

Validation, limitations, and open decisions: Reject unsupported tokens and malformed syntax with a location-aware message. Function list, implicit multiplication, exponent syntax, angle units, and numeric limits are not defined in the source and need agreement.

Dependencies and risks: Feeds the calculation engine (ID 2), input diagnostics (ID 11), step breakdown (ID 10), algebra (ID 20), and graph/calculus inputs (IDs 6, 28, 30). Risk: parser and engine conventions can drift.

Proposed definition of done: Team-approved grammar is documented; listed valid/invalid expression tests pass; precedence and function behavior are reproducible; diagnostics and evidence are attached.

Requirement 01 – Test Evidence

Implementation: Expression parser supporting arithmetic operations, operator precedence, parentheses, scientific functions, and syntax error detection.

Test Date: November 8, 2026 (by Ivan Kullaya)

Test Environment: Web browser (Google Chrome)

Test ID	Test Performed	Expected result	Status
R01-T1	2 + 3 x 4	14	Pass
R01-T2	(2 + 3) x 4	20	Pass
R01-T3	sin(π/2)	1	Pass
R01-T4	2 + (3 x 4)	14	Pass
R01-T5	2 + 3 x	Error identifying incomplete expression	Pass
R01-T6	2 @ 3	Error identifying unsupported character	Pass
R01-T7	2+3×4 and 2 + 3 × 4	Both return 14	Pass

Evidence: GitHub R01 Evidence Folder

Implementation Details: The parser uses tokenization and recursive parsing to interpret mathematical expressions. It supports addition, subtraction, multiplication, division, exponentiation, parentheses, and scientific functions such as sin, cos, tan, sqrt, ln, and log. It also supports implicit multiplication and recognizes invalid expressions with error messages identifying the location of the issue.

Result: All four tests passed, confirming that Requirement 01 meets its acceptance criteria.

Requirement 02 — Arbitrary-precision calculation engine

Proposed / hypothetical • Story points: 1 • Priority: 2 • Sprint: 1

Source description: A basic arbitrary-precision calculation engine ensuring accurate decimal results without standard floating-point rounding errors.

What it should do

Evaluate parsed numeric operations with a representation and precision policy that avoid ordinary binary floating-point surprises for decimal inputs.

Acceptance criteria

- Decimal operations preserve the intended decimal result under the approved precision policy.
- Displayed rounding does not silently change the engine's stored value.
- Overflow, division by zero, and precision-limit behavior are defined and reported safely.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R02-T1	0.1 + 0.2	0.3 exactly under the proposed decimal semantics.
R02-T2	1 ÷ 3, then multiply the stored result by 3	1 within the documented internal precision policy; the display may round separately.

UI / workflow notes: Show the evaluated result in the shared result area; if precision limits apply, make them discoverable without replacing the input.

Validation, limitations, and open decisions: The source does not specify precision, rounding mode, exponent range, or whether arbitrary precision is exact rational, decimal, or configurable. Agree these limits before final acceptance.

Dependencies and risks: Consumes the parser output (ID 1) and supports most numeric features. Risk: using display-rounded values in later calculations would violate the preservation goal in ID 14.

Proposed definition of done: The numeric representation and limits are documented; decimal and precision-boundary cases pass; downstream features use the full stored result; evidence records actual outputs.

Requirement 02 – Test Evidence

Implementation: Arbitrary-precision calculation engine intended to support accurate decimal arithmetic, internal precision preservation, and safe handling of mathematical errors.

Test Date: November 9, 2026 (by Ivan Kullaya)

Test Environment: Web browser (Google Chrome)

Test ID	Test Performed	Expected result	Status
R02-T1	Calculate 0.1 + 0.2	Result is exactly 0.3 under the approved decimal precision policy	Pass
R02-T2	Calculate 1 ÷ 3, then multiply the stored result by 3	Result is 1 within the documented internal precision policy	Fail / Needs Review
R02-T3	Calculate 10 ÷ 0	Division-by-zero error message appears without crashing the calculator	Pass
R02-T4	Enter a calculation exceeding the supported numeric range	Calculator safely reports an overflow or precision-limit error	Pass
R02-T5	Calculate 0.1 + 0.2 + 0.3	Result is 0.6 under the approved decimal precision policy	Pass

Evidence: GitHub R02 Evidence Folder

Implementation Details:

The calculation engine is intended to evaluate mathematical expressions using a numeric representation that minimizes or eliminates standard floating-point rounding errors.

The implementation must define its internal numeric representation, supported precision, rounding rules, and numeric limits. Display formatting should remain separate from internal calculations so that subsequent operations use the stored result rather than a rounded display value.

Division by zero, overflow, and unsupported precision cases should produce appropriate error messages without interrupting the calculator's functionality.

Result: Requirement 02 is partially verified. Additional testing and potential improvements to internal precision handling are needed before the requirement can be marked complete.

Requirement 03 — Persistent session history and Ans recall

Proposed / hypothetical • Story points: 1 • Priority: 3 • Sprint: 1

Source description: A simple persistent session history allowing users to recall past inputs and outputs via a standard "Ans" key.

What it should do

Keep a chronological list of calculations and results that survives the agreed persistence boundary, and let the next expression refer to the previous result through Ans.

Acceptance criteria

- A completed calculation appears with its original input and result in history.
- Ans resolves to the prior result, not the formatted history label.
- History remains available after the persistence event selected by the team, such as app restart.

Example test cases — proposed, not run

Test ID	Input / action	Expected output	Status
R03-T1	Evaluate 2 + 3, then Ans + 4	History contains 2 + 3 → 5; second result is 9.	Not run
R03-T2	Evaluate 7 × 6, close and reopen the app	History contains the saved 7 × 6 → 42 if restart persistence is approved.	Not run

UI / workflow notes: History list shows expression and result; selecting an item may recall its expression. Keep Ans available in the expression workflow.

Validation, limitations, and open decisions: Define what counts as a session, history retention/clear behavior, ordering, and whether Ans resets after errors or mode changes. Avoid saving incomplete expressions as completed results.

Dependencies and risks: Uses calculation results from ID 2 and persistence storage. Mode preservation (ID 7), favorites (ID 21), export (ID 23), and notes (ID 33) may reuse history records.

Proposed definition of done: History and Ans rules are agreed; recall and persistence cases pass; ordering and error behavior are documented; evidence shows saved records and actual results.

Requirement 03 – Test Evidence

Implementation: Persistent calculation history and Ans recall using JavaScript and localStorage.

Test Date: November 8, 2026 (by Ivan Kullaya)

Test Environment: Web browser (Google Chrome)

Test ID	Test Performed	Expected result	Status
R03-T1	Calculate 2 + 3, then Ans + 4	Results are 5 and 9; calculations appear in history	Pass
R03-T2	Calculate 7 × 6, then close and reopen the calculator	History still contains 7 × 6 = 42	Pass
R03-T3	Select a previous calculation from history	Previous expression is restored to the input field	Pass
R03-T4	Clear calculation history	Saved history is removed	Pass

Evidence: GitHub R03 Evidence Folder

Implementation Details: Calculation history is stored using browser localStorage. Each successful calculation saves the original expression and numerical result. The Ans variable stores the previous successful result for use in future calculations. History is limited to the 100 most recent calculations.

Result: All four test passed, confirming that Requirement 03 meets its acceptance criteria.

Requirement 04 — Unit conversion

Proposed / hypothetical • Story points: 2 • Priority: 4 • Sprint: 1

Source description: A standard conversion module handling the most common length, weight, volume, temperature, and basic currency units.

What it should do

Convert a value between team-approved common units in length, mass/weight, volume, temperature, and basic currencies, showing the source and converted values with units.

Acceptance criteria

- Conversions use the correct category and unit-specific formula.
- Temperature conversions apply offsets as well as scale factors.
- Currency conversions identify the rate and its date/source when rates are time-dependent.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R04-T1	1 m → cm	100 cm.
R04-T2	0 °C → °F	32 °F.
R04-T3	$10 USD → EUR	Converted amount = 10 × the selected USD→EUR rate; show rate/date/source. Numeric result depends on the team-approved rate source and timestamp.

UI / workflow notes: Provide value, source-unit, destination-unit, and converted-result fields; group units by category and make the active category clear.

Validation, limitations, and open decisions: Reject incompatible units and non-finite values. The supported unit list, currency-rate provider, offline behavior, rate timestamp, and rounding are not specified; agree them before final acceptance.

Dependencies and risks: Needs unit definitions and a conversion engine; currency needs an agreed external or stored rate source. Risk: stale currency rates can make a plausible but misleading result.

Proposed definition of done: Approved unit lists/formulas and currency policy are documented; length, temperature, invalid-category, and rate-attribution cases pass; evidence includes actual rate metadata.

Requirement 04 – Test Evidence

Implementation: Unit conversion module supporting length, weight, volume, temperature, and currency conversions with selectable source and destination units.

Test Date: November 8, 2026 (by Ivan Kullaya)

Test Environment: Web browser (Google Chrome)

Test ID	Test Performed	Expected result	Status
R04-T1	Convert 1 meter to centimeters	100 cm	Pass
R04-T2	Convert 0°C to Fahrenheit	32°F	Pass
R04-T3	Convert 10 USD to EUR	Correct converted amount using the selected exchange rate, with rate, date, and source displayed	Pass
R04-T4	Convert 5 kilograms to grams	5,000 g	Pass
R04-T5	Convert 2 liters to milliliters	2,000 mL	Pass
R04-T6	Attempt conversion between incompatible units	Error message or prevention of invalid conversion	Pass

Evidence: GitHub R04 Evidence Folder

Implementation Details:

The unit conversion module is intended to provide conversions across five categories: length, weight, volume, temperature, and currency. Users select a conversion category, enter a numerical value, choose the source and destination units, and receive the converted result.

Length, weight, and volume conversions use predefined conversion factors. Temperature conversions use formulas that account for both scale and offset. Currency conversions require an approved exchange-rate source and should display the rate and its associated date.

The module should also prevent incompatible conversions and reject invalid numerical inputs.

Result: All four tests passed, confirming that Requirement 04 meets its acceptance criteria.

Requirement 05 — Clean, user-friendly UI layout

Proposed / hypothetical • Story points: 2 • Priority: 5 • Sprint: 1

Source description: A clean, user-friendly UI layout.

What it should do

Present the calculator's primary input, result, key actions, and available modes in a readable layout that users can operate without guessing where results or controls are.

Acceptance criteria

- Input, result, and primary actions have clear labels and visible focus/selection.
- A new user can complete the agreed basic calculation task without an instruction from the tester.
- Layout remains usable at the team-approved window sizes and text scaling.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R05-T1	Ask a first-time tester to enter 2 + 3 and find the result	Tester completes the task and identifies 5 without coaching; record task steps and issues.
R05-T2	Resize to the smallest supported window and increase text size	Primary input, result, and actions remain visible or reachable without overlap.

UI / workflow notes: This requirement is the overall layout foundation. Record target platforms/window sizes and keyboard-accessible paths in the review notes.

Validation, limitations, and open decisions: "Clean" and "user-friendly" are subjective until usability targets, accessibility expectations, and supported sizes are agreed.

Dependencies and risks: Provides the shared shell for all modes and features (IDs 7, 8, 22, 34). Risk: feature-specific controls may overcrowd the primary workflow.

Proposed definition of done: Team-approved usability checks pass on target sizes; labels, focus, and navigation are verified; tester observations and screenshots are attached.

Requirement 05 – Test Evidence

Implementation: Persistent calculation history and Ans recall using JavaScript and localStorage.

Test Date: November 8, 2026 (by Ivan Kullaya)

Test Environment: Web browser (Google Chrome)

Test ID	Test Performed	Expected result	Status
R05-T1	First-time user enters 2 + 3 and presses =	User completes calculation and identifies result 5 without assistance	Pass
R05-T2	Resize browser to smallest supported window	Calculator buttons, input, and result remain accessible without overlap	Pass
R05-T3	Resize browser to smallest supported window	Interactive controls have visible focus indicators	Pass
R05-T4	Increase browser text scaling to 200%	Calculator remains readable and functional without overlapping controls	Pass

Evidence: GitHub R05 Evidence Folder

Implementation Details: The calculator interface is built using HTML and CSS with a centered layout, organized button grids, and separate input and result displays. Scientific functions are grouped separately from basic arithmetic controls. CSS Grid maintains consistent button spacing, while responsive sizing allows the calculator to adapt to different browser widths. The interface also includes visible keyboard focus indicators, light and dark color schemes, and accessible labels for important controls.

Result: All four tests passed, confirming that Requirement 05 meets its acceptance criteria.

Requirement 06 — 2D single-variable graphing

Proposed / hypothetical • Story points: 2 • Priority: 6 • Sprint: 1

Source description: A 2D graphing module, able to plot single variable equations on a canvas viewer with controls such as pan and zoom.

What it should do

Plot a selected single-variable function on a two-dimensional coordinate plane and let users move and scale the visible graph.

Acceptance criteria

- A valid function is plotted over a visible x-range with correctly placed axes.
- Pan changes the displayed coordinate window without changing the function.
- Zoom changes the displayed scale and retains the plotted function.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R06-T1	Plot y = 2x over a range containing 0 and 1	Line passes through (0, 0) and (1, 2), within the agreed rendering tolerance.
R06-T2	Pan right, then zoom in once	Viewport center/range changes as indicated by the controls; plotted line remains y = 2x.

UI / workflow notes: Show graph canvas, coordinate axes/ticks, function identity, and discoverable pan/zoom controls. State the active range.

Validation, limitations, and open decisions: Define supported functions, sampling/tolerance, discontinuity handling, default ranges, and pan/zoom increments. A graph is a visual approximation, not proof of symbolic correctness.

Dependencies and risks: Uses expression syntax from ID 1 and a graph renderer; customization (ID 19), comparison (ID 24), intercepts (ID 30), and PNG export (ID 32) build on this view.

Proposed definition of done: Approved function set and plotting tolerance are documented; known-point and viewport tests pass; screenshots show axes, correct plot, and pan/zoom state.

Requirement 06 – Test Evidence

Implementation: Persistent calculation history and Ans recall using JavaScript and localStorage.

Test Date: November 8, 2026 (by Ivan Kullaya)

Test Environment: Web browser (Google Chrome)

Test ID	Test Performed	Expected result	Status
R06-T1	Plot y = 2x over a range containing 0 and 1	Graph displays a straight line passing through (0, 0) and (1, 2)	Blocked — Unable to Test
R06-T2	Pan the graph to the right	Viewport shifts horizontally while preserving the plotted function	Blocked — Unable to Test
R06-T3	Zoom in on the graph	Graph scale changes while maintaining the correct function	Blocked — Unable to Test
R06-T4	Plot $y = x^2$	Graph displays a parabola with its vertex at (0, 0)	Blocked — Unable to Test
R06-T5	Zoom out and return to the original viewing range	Graph remains visible with correctly positioned axes and coordinates	Blocked — Unable to Test

Evidence: GitHub R06 Evidence Folder

Implementation Details:

The 2D graphing module is intended to allow users to enter single-variable equations and display their corresponding graphs on a coordinate plane. The module should support interactive controls for panning and zooming while maintaining the mathematical accuracy of the plotted function.

The graphing functionality is currently experiencing implementation issues that prevent the planned tests from being completed. As a result, the graphing feature has not yet been verified against its acceptance criteria.

The test cases have been documented so they can be executed once the functionality is restored.

Result: Requirement 06 is not yet verified. Testing is currently blocked because the graphing module requires additional fixes. All planned test cases have been documented but remain unexecuted.

Once the graphing functionality is restored, the assigned developer can complete the tests, record the actual results, and attach screenshots demonstrating successful plotting, panning, and zooming.

Requirement 07 — Calculation mode switcher

Proposed / hypothetical • Story points: 1 • Priority: 7 • Sprint: 1

Source description: A calculation mode switcher that allows users to choose between basic, scientific, financial, and conversion modes. Current input/session history should be preserved.

What it should do

Switch among basic, scientific, financial, and conversion modes while retaining the current draft input and the existing session history.

Acceptance criteria

- Each named mode can be selected and is visibly identified.
- Switching modes does not erase or silently alter the current input.
- Existing session history remains accessible after a mode change.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R07-T1	Enter draft 12 × (3 + 1), switch Basic → Scientific → Basic	Draft remains exactly 12 × (3 + 1), unless team-approved mode validation reports it unsupported.
R07-T2	Create history item 2 + 3 → 5, switch to Conversion	History item remains available and unchanged.

UI / workflow notes: Use a persistent mode selector and visibly indicate the active mode; distinguish a draft from a completed history entry.

Validation, limitations, and open decisions: Define behavior for expressions unsupported in the destination mode, and whether mode-specific settings persist independently.

Dependencies and risks: Requires the shared UI (ID 5), mode features (IDs 1, 4, 9), and session history (ID 3). Risk: mode switch may re-interpret rather than preserve input.

Proposed definition of done: Mode names and transition behavior are agreed; draft and history preservation tests pass; screenshots demonstrate each mode and state retention.

Requirement 07 – Test Evidence

Implementation: Calculation mode switcher allowing users to navigate between Basic, Scientific, Financial, and Conversion modes while preserving current input and calculation history.

Test Date: November 9, 2026 (by Ivan Kullaya)

Test Environment: Web browser (Google Chrome)

Test ID	Test Performed	Expected result	Status
R07-T1	Enter 12 × 3 + 1, switch Basic → Scientific → Basic	Original expression remains unchanged	Pass
R07-T2	Calculate 2 + 3 = 5, then switch to Conversion mode	Previous calculation remains available in history	Pass
R07-T3	Switch between Basic, Scientific, Financial, and Conversion modes	Each mode loads correctly and is visibly identified	Fail
R07-T4	Enter an unfinished expression, then switch modes	Draft expression is preserved without being cleared or modified	Pass
R07-T5	Switch modes multiple times and return to the original mode	Current input and saved calculation history remain accessible	Partial Pass

Evidence: GitHub R07 Evidence Folder

Implementation Details:

The calculation mode switcher is intended to provide navigation between Basic, Scientific, Financial, and Conversion modes using a shared calculator interface.

Each mode should be clearly identified through the mode selector or active mode indicator. Switching modes should preserve the current expression without silently changing its contents.

Calculation history should remain available across mode changes, allowing users to access previously completed calculations regardless of the active mode.

The implementation should also define how unsupported expressions are handled when switching to a mode that does not support certain functions.

Result: Requirement 02 is partially verified. The mode switcher successfully preserves user input and calculation history. However, improvements are needed to ensure each calculation mode displays its corresponding interface.

The Financial and Conversion modes should display their respective tools when selected, rather than requiring users to access them through secondary buttons in Scientific mode. Additionally, secondary panels should close or hide automatically when switching to a different mode.

These issues should be addressed and retested before Requirement 07 is considered fully complete.

Requirement 08 — Function search and category filter

Proposed / hypothetical • Story points: 1 • Priority: 8 • Sprint: 1

Source description: A category filter system that allows the user to search for functions by category.

What it should do

Let users find calculator functions by text and by a category label such as the team-approved algebra, statistics, finance, or trigonometry groups.

Acceptance criteria

- Choosing a category limits results to functions tagged with that category.
- Search narrows the visible functions using an agreed name/description matching rule.
- No-match results are clearly communicated and do not appear as valid functions.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R08-T1	Select Trigonometry and search “sin”	sin appears if it is in the approved function catalog; unrelated functions are excluded.
R08-T2	Search for “no_such_function”	An empty-state message appears; no unrelated result is shown.

UI / workflow notes: Provide a category selector, search field, result list, and a clear empty state; selecting a result should reveal its accepted syntax.

Validation, limitations, and open decisions: The function catalog, category tags, case sensitivity, aliases, and whether search includes descriptions need agreement.

Dependencies and risks: Depends on a maintained function catalog and UI (ID 5); parser support (ID 1) must match selectable functions. Risk: search advertises unsupported syntax.

Proposed definition of done: Approved catalog and category map are reviewed; category, matching, and empty-state cases pass; evidence records returned functions and syntax.

Requirement 08 – Test Evidence

Implementation: Function search and category filtering system allowing users to locate calculator functions by category or search term.

Test Date: November 8, 2026 (by Ivan Kullaya)

Test Environment: Web browser (Google Chrome)

Test ID	Test Performed	Expected result	Status
R08-T1	Select Trigonometry category and search sin	Matching trigonometric functions appear; unrelated functions are excluded	Pass
R08-T2	Search for no_such_function	No matching functions message appears	Pass
R08-T3	Select a function category without entering a search term	Only functions belonging to the selected category are displayed	Pass
R08-T4	Select a function from the filtered results	Selected function displays its accepted syntax or inserts it into the expression field	Pass

Evidence: GitHub R08 Evidence Folder

Implementation Details:

The function search and category filter system is designed to organize available calculator functions into categories such as trigonometry, algebra, statistics, and finance, depending on the team's approved function catalog.

Users can select a category to narrow the displayed functions and enter keywords to search for specific functions. The search system should display only functions matching the selected category and search term.

When no matching functions exist, the interface should display a clear message rather than unrelated results. Selecting a function should provide its accepted syntax to help users enter valid expressions.

Result: All four tests passed, confirming that Requirement 08 meets its acceptance criteria.

Requirement 09 — Financial calculations

Proposed / hypothetical • Story points: 2 • Priority: 9 • Sprint: 2

Source description: A standard financial calculation module covering compound interest, loan payments, etc.

What it should do

Calculate at least compound-interest growth and periodic loan payments using explicitly labeled financial inputs and assumptions.

Acceptance criteria

- Compound-interest results follow the selected rate/compounding convention.
- Loan payment results use the selected payment timing, rate period, and number of payments.
- Inputs and outputs show units/periods, with currency rounded only for display according to policy.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R09-T1	Principal $1,000; nominal annual rate 10%; monthly compounding; 1 year	Future value $1,104.71 to cents, using A=P(1+0.10/12)^12.
R09-T2	Loan $10,000; 6% nominal APR; monthly payments; 36 months; end-of-month payments	$304.22 monthly to cents, using monthly rate 0.06/12 and the standard amortizing-payment formula.

UI / workflow notes: Use labeled fields for principal, rate, term, compounding/payment frequency, and result. Show the selected assumptions beside the result.

Validation, limitations, and open decisions: Reject negative term, nonpositive payment count, invalid rate formats, and missing units. The source does not define APR vs effective rate, payment timing, fees, taxes, or rounding; examples above propose nominal annual rates and monthly periods.

Dependencies and risks: Needs numeric engine (ID 2), finance formulas, and possibly scenario comparison (ID 25). Risk: unstated conventions can create materially different answers.

Proposed definition of done: Finance conventions and formulas are team-approved; both examples and invalid-input cases pass; displayed assumptions and actual outputs are captured.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 10 — Calculation-step and formula breakdown

Proposed / hypothetical • Story points: 2 • Priority: 10 • Sprint: 2

Source description: A breakdown view that shows the calculation steps, including the formula definitions when needed.

What it should do

Explain how a supported calculation reaches its result by showing ordered operations and, for named formulas, the formula definition with substituted values.

Acceptance criteria

- Steps are ordered consistently with the evaluator and precedence rules.
- The final displayed step agrees with the calculator result.
- A formula-based feature can show its selected formula and input substitution.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R10-T1	Request a breakdown for 2 + 3 × 4	Steps show 3 × 4 = 12, then 2 + 12 = 14.
R10-T2	Request the compound-interest breakdown for the R09-T1 inputs	Show A=P(1+r/n)^(nt), substitute the agreed values, and end at $1,104.71 to cents.

UI / workflow notes: Place a clearly labeled expandable steps view near the result; keep formula symbols and units readable.

Validation, limitations, and open decisions: Define which features must provide steps and how to handle operations whose internal algorithm is not naturally a human-readable sequence.

Dependencies and risks: Must use parser/evaluator semantics (IDs 1–2) and feature formulas (for example ID 9). Risk: a separately generated explanation can disagree with the actual computation.

Proposed definition of done: Breakdown is generated from the calculation's real operation/formula record; step order and final result match; representative evidence is attached.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 11 — Helpful input checker

Proposed / hypothetical • Story points: 1 • Priority: 11 • Sprint: 2

Source description: An input checker with helpful explanations for issues like missing parentheses or division by zero.

What it should do

Detect common malformed or invalid expressions and explain the problem in terms the user can act on.

Acceptance criteria

- A missing parenthesis is identified with a corrective message.
- Division by zero is reported as an error, not a valid numeric result.
- An invalid expression does not crash the app or overwrite the prior valid result without explanation.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R11-T1	Evaluate 2 + (3	Show an unmatched/missing closing-parenthesis message; do not claim a result.
R11-T2	Evaluate 8 ÷ 0	Show a division-by-zero message; do not return infinity as an ordinary result.

UI / workflow notes: Place a concise message near the input; indicate the relevant token/location when available and preserve editable input for correction.

Validation, limitations, and open decisions: Agree error categories, location conventions, localization, and whether the last successful result remains visible but marked as previous.

Dependencies and risks: Depends on parser and numeric engine (IDs 1–2). Risk: duplicated error rules may disagree with actual evaluation.

Proposed definition of done: Each supported error has a reproducible trigger and actionable message; no crash or misleading result occurs; screenshots/logs of actual diagnostics are attached.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 12 — Undo and redo for expression editing

Proposed / hypothetical • Story points: 1 • Priority: 12 • Sprint: 2

Source description: Undo and redo for expression editing.

What it should do

Let users reverse and reapply expression edits in the active editing context without undoing unrelated calculations or settings.

Acceptance criteria

- Undo restores the previous expression state.
- Redo reapplies the state that was undone.
- A new edit after undo follows the agreed redo-history clearing rule.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R12-T1	Type 2 + 3, change 3 to 4, Undo	Editor returns to 2 + 3.
R12-T2	Redo after R12-T1	Editor returns to 2 + 4.

UI / workflow notes: Provide discoverable undo/redo controls and keyboard shortcuts if agreed; disabled state should be clear when no action is available.

Validation, limitations, and open decisions: Define history depth and whether undo applies only to text edits, clipboard changes, or other editor actions.

Dependencies and risks: Uses the expression editor (ID 5) and mode behavior (ID 7). Risk: shortcut handling can conflict with app/browser/system conventions.

Proposed definition of done: Editing history follows the approved scope; undo/redo and branch-after-undo cases pass; the restored expression is visible and evidence is attached.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 13 — Number theory tools

Proposed / hypothetical • Story points: 3 • Priority: 13 • Sprint: 2

Source description: Number theory tools: prime check, prime factorization, greatest common divisor, and least common multiple.

What it should do

Provide prime testing, integer factorization, greatest common divisor, and least common multiple for supported integer inputs.

Acceptance criteria

- Prime check distinguishes primes and composites for the supported integer domain.
- Factorization multiplies back to the original positive integer.
- GCD and LCM return mathematically consistent integer results.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R13-T1	Prime check 17 and 21	17 is prime; 21 is composite.
R13-T2	Factorize 60	$2^2 \times 3 \times 5$.
R13-T3	GCD(12,18), LCM(12,18)	6 and 36, respectively.

UI / workflow notes: Expose each operation with clearly labeled integer inputs and result format; show factors as a product with exponents or an agreed equivalent.

Validation, limitations, and open decisions: Define behavior for 0, 1, negative values, non-integers, and very large integers. Factorization runtime can grow quickly for large inputs.

Dependencies and risks: Uses integer arithmetic (ID 2) and function catalog/search (ID 8). Risk: edge-case definitions, especially factorization of zero/negative values, are absent.

Proposed definition of done: Input domain and edge cases are agreed; prime, factorization, GCD, and LCM cases pass, including selected boundaries; actual outputs are recorded.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 14 — Displayed precision controls

Proposed / hypothetical • Story points: 2 • Priority: 14 • Sprint: 2

Source description: Precision controls for displayed decimal places or significant figures while preserving the full stored value

What it should do

Let users choose decimal-place or significant-figure formatting while retaining the unrounded calculation value for subsequent operations.

Acceptance criteria

- The chosen display mode and precision are applied consistently to results.
- Changing display precision does not mutate the stored result.
- Subsequent calculations use the full stored value, not the rounded text.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R14-T1	Calculate 1 ÷ 3; show 2 decimal places, then 4	Display changes from 0.33 to 0.3333.
R14-T2	After calculating 1 ÷ 3, calculate Ans × 3	Result is 1 within the documented engine precision, regardless of the earlier display formatting.

UI / workflow notes: Offer a labeled precision control and make the active decimal-place/significant-figure mode visible.

Validation, limitations, and open decisions: Agree rounding mode, significant-figure handling for zero, maximum precision, and how precision is represented in exports/history.

Dependencies and risks: Depends on numeric engine (ID 2), history/Ans (ID 3), and formatting shared by financial/statistical results. Risk: rounded display text may be reused as data.

Proposed definition of done: Formatting rules are approved; visual formatting and full-value reuse tests pass; evidence shows both displayed forms and the unchanged computed value.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 15 — Tolerance calculator

Proposed / hypothetical • Story points: 2 • Priority: 15 • Sprint: 2

Source description: A tolerance calculator that computes minimum and maximum results from measurements with specified tolerances.

What it should do

For a measurement with a nominal value and tolerance, report the allowable minimum and maximum under the agreed tolerance-combination rules.

Acceptance criteria

- For a single symmetric tolerance, minimum = nominal − tolerance and maximum = nominal + tolerance.
- Units are retained or converted consistently.
- Invalid negative tolerance is rejected or explained according to the approved rule.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R15-T1	Length 10.0 mm ± 0.2 mm	Minimum 9.8 mm; maximum 10.2 mm.
R15-T2	Nominal 5, tolerance 0	Minimum 5; maximum 5.

UI / workflow notes: Provide nominal value, tolerance, and unit inputs; display the interval as [minimum, maximum] and show units.

Validation, limitations, and open decisions: Define supported measurement operations and how tolerances combine for sums, products, correlated variables, asymmetric tolerances, and rounding.

Dependencies and risks: Uses numeric arithmetic and possibly unit conversion (IDs 2 and 4). Risk: independent worst-case intervals may not reflect correlated measurements.

Proposed definition of done: The supported tolerance model is documented; boundary and unit cases pass; interval values and assumptions are visible in captured evidence.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 16 — Statistical calculations

Proposed / hypothetical • Story points: 3 • Priority: 16 • Sprint: 2

Source description: A statistical calculation module that allows users to calculate mean, median, mode, standard deviation, and variance for a given dataset.

What it should do

Accept a dataset and compute the five listed descriptive statistics using a documented population/sample convention.

Acceptance criteria

- Mean, median, mode, standard deviation, and variance are calculated from the same validated dataset.
- Empty and malformed datasets produce an explanatory result instead of a numeric claim.
- The variance convention and behavior when multiple/no modes exist are visible or documented.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R16-T1	Dataset [1, 2, 2, 3]; proposed population convention	Mean 2; median 2; mode 2; population variance 0.5; population standard deviation ≈0.7071.
R16-T2	Dataset []	Explain that statistics cannot be computed for an empty dataset.
R16-T3	Dataset [1, 2, 3, 4]; proposed population convention	Mean 2.5; median 2.5; population variance 1.25; standard deviation ≈1.1180; mode behavior must follow the team's selected rule.

UI / workflow notes: Offer a dataset entry workflow (for example comma-separated values) and display each statistic with a clear label.

Validation, limitations, and open decisions: The source does not choose sample vs population variance, mode behavior for ties/no mode, accepted separators, or missing-value handling. Tests above propose population variance.

Dependencies and risks: Needs numeric engine (ID 2), dataset parser, and precision formatting (ID 14). Risk: an undocumented convention produces a numerically plausible but incompatible answer.

Proposed definition of done: Dataset syntax and all statistical conventions are approved; known-value, tie/no-mode, empty, and invalid-input cases pass; actual outputs are recorded.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 17 — Fraction calculator

Proposed / hypothetical • Story points: 3 • Priority: 17 • Sprint: 2

Source description: A fraction calculator that allows users to perform arithmetic operations with fractions and convert between fractions and decimals.

What it should do

Perform the agreed arithmetic operations on fractions and convert between fractional and decimal representations.

Acceptance criteria

- Fraction arithmetic returns an equivalent correctly reduced fraction under the approved formatting rule.
- Fraction/decimal conversion preserves value within the selected decimal precision.
- Zero denominators are rejected with a clear message.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R17-T1	1/2 + 1/3	5/6.
R17-T2	0.125 → fraction	1/8.
R17-T3	1/0	Clear zero-denominator error; no numeric result.

UI / workflow notes: Allow fraction entry and show both exact fraction and optional decimal value where useful.

Validation, limitations, and open decisions: Define mixed-number and negative syntax, reduction rules, recurring decimal handling, and precision for decimal-to-fraction conversion.

Dependencies and risks: Uses parser and numeric/rational arithmetic (IDs 1–2), and may share display precision rules (ID 14). Risk: decimal-to-fraction conversion may have many valid approximations.

Proposed definition of done: Fraction syntax and reduction policy are approved; arithmetic, conversion, and denominator errors pass; exact and displayed forms are captured.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 18 — Percentage calculator

Proposed / hypothetical • Story points: 2 • Priority: 18 • Sprint: 2

Source description: A percentage calculator that allows users to calculate percentage increases, decreases, and differences between values.

What it should do

Calculate percentage increase, decrease, and difference using explicitly labeled starting and comparison values.

Acceptance criteria

- Increase and decrease use the starting value as the denominator.
- The result states the direction and percent unit.
- Percentage-difference denominator is defined by an approved convention.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R18-T1	Increase from 80 to 100	25% increase.
R18-T2	Decrease from 100 to 80	20% decrease.
R18-T3	Difference between 80 and 100; proposed symmetric convention	22.22% = |100−80| / ((100+80)/2) × 100, rounded to 2 decimals.

UI / workflow notes: Offer separate labeled workflows for increase, decrease, and difference so the baseline is unambiguous.

Validation, limitations, and open decisions: Define percentage-difference convention, behavior for zero/negative baselines, and rounding. R18-T3 is a proposal that requires team agreement.

Dependencies and risks: Uses numeric engine and display formatting (IDs 2 and 14). Risk: users may mistake symmetric percentage difference for percent increase from a baseline.

Proposed definition of done: Each workflow labels its denominator/convention; examples and zero-baseline cases follow the documented rule; evidence shows direction and units.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 19 — Graph customization

Proposed / hypothetical • Story points: 3 • Priority: 19 • Sprint: 2

Source description: A graph customization feature that allows users to adjust axis ranges, gridlines, and graph labels.

What it should do

Let users set visible axis ranges, show or hide gridlines, and edit graph labels without changing the underlying plotted equations.

Acceptance criteria

- Valid manual axis bounds are reflected in the viewport.
- Gridline visibility follows the selected setting.
- Edited labels appear on the graph and remain associated with the intended axis/graph.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R19-T1	Plot y=x; set x and y ranges to −2…2	Both axes show the selected bounds and the line is clipped/displayed within that viewport.
R19-T2	Turn grid off, rename graph to “y=x demo”	Gridlines disappear and the graph label reads “y=x demo”.

UI / workflow notes: Put range, gridline, and label controls near the graph; show when manual ranges override automatic scaling.

Validation, limitations, and open decisions: Reject nonnumeric bounds and min ≥ max; define range units, tick spacing, and whether custom settings persist per graph.

Dependencies and risks: Builds on graphing (ID 6) and supports graph comparison/export (IDs 24 and 32). Risk: independent settings across multiple graphs can be unclear.

Proposed definition of done: Axis, grid, and label rules are approved; valid/invalid controls update only the intended graph; screenshots show each configured state.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 20 — Basic algebra equation solver

Proposed / hypothetical • Story points: 5 • Priority: 20 • Sprint: 2

Source description: An equation solver that allows users to enter basic algebraic equations and solve for an unknown variable.

What it should do

Parse a team-approved class of basic algebra equations and return the solution for the requested unknown, including clear no-solution or multiple-solution outcomes where applicable.

Acceptance criteria

- A solvable supported equation returns a value satisfying the original equation.
- The solution is associated with the correct variable.
- Inconsistent and identity equations are distinguished if included in the supported scope.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R20-T1	Solve 2x + 3 = 11 for x	x = 4; substitution gives 11 = 11.
R20-T2	Solve x = x + 1	No solution.
R20-T3	Solve x = x	Every allowed x is a solution (identity), if identities are in the approved scope.

UI / workflow notes: Show the equation, selected unknown variable, and solution status; preserve the original equation for verification.

Validation, limitations, and open decisions: Define supported degree, variable syntax, implicit multiplication, domains, and treatment of nonlinear/multiple/complex solutions. “Basic” is not a precise equation grammar.

Dependencies and risks: Needs expression parser (ID 1), symbolic or equation-solving logic, and numeric engine (ID 2). Risk: unsupported equations may be mistaken for unsolvable equations.

Proposed definition of done: Equation class and solution semantics are agreed; solvable, inconsistent, identity, and unsupported cases are correctly distinguished; evidence includes substitution checks.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 21 — Favorite calculations

Proposed / hypothetical • Story points: 2 • Priority: 21 • Sprint: 2

Source description: A favorite systems that lets users save frequently used calculations for quick access.

What it should do

Let users save a calculation to a favorites collection and retrieve it later without retyping its expression.

Acceptance criteria

- A user can add and identify a favorite calculation.
- Selecting a favorite restores the saved expression or calculation using the approved recall behavior.
- Saved favorites remain available across the agreed persistence boundary.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R21-T1	Save 2 + 3 as “Simple sum”, then reopen favorites	“Simple sum” appears with its expression; selecting it restores 2 + 3 and its result is 5 if results are stored.
R21-T2	Save the same expression twice	Duplicate handling follows the selected policy; no silent overwrite occurs.

UI / workflow notes: Provide a favorite action and a list with descriptive names; make recall distinct from immediate execution if that is the chosen behavior.

Validation, limitations, and open decisions: Define naming, duplicate handling, edit/delete, storage scope, and whether a favorite stores an expression, result, mode, or all of them.

Dependencies and risks: May reuse history records (ID 3), persistence, and mode context (ID 7). Risk: recalled calculations may depend on a changed rate or setting.

Proposed definition of done: Saved fields and recall semantics are agreed; save/recall/duplicate cases pass; evidence shows the item in the list and its restored calculation.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 22 — Display customization

Proposed / hypothetical • Story points: 2 • Priority: 22 • Sprint: 2

Source description: Customization that lets users change colors, labels, and display settings.

What it should do

Let users adjust approved visual settings and labels, then apply those choices consistently to the relevant calculator views.

Acceptance criteria

- A supported color or label setting changes the intended view.
- A display setting changes its documented presentation behavior.
- Settings do not alter stored calculation values.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R22-T1	Change a graph/accent color and label	The selected color and label appear in the target view.
R22-T2	Change a display setting, then evaluate 2 + 3	Presentation follows the setting and the numeric result remains 5.

UI / workflow notes: Provide a settings area with named controls, current values, and a way to restore defaults if approved.

Validation, limitations, and open decisions: Define which colors/labels/display settings are customizable, persistence, allowed ranges, and accessibility contrast requirements.

Dependencies and risks: Uses shared theme/display tokens and UI (ID 5); may overlap with graph labels (ID 19) and light/dark mode (ID 34). Risk: settings may conflict or apply inconsistently.

Proposed definition of done: Customization scope and persistence are agreed; changed/default settings render correctly without affecting calculations; screenshots and setting values are captured.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 23 — Export calculation history

Proposed / hypothetical • Story points: 3 • Priority: 23 • Sprint: 2

Source description: An export tool that lets users download their calculation history as a text or CSV.

What it should do

Export selected or complete calculation history as a text file or CSV that preserves the agreed record fields and can be read outside the app.

Acceptance criteria

- Text and CSV exports contain the expected history entries.
- CSV fields remain correctly delimited/escaped for commas, quotes, and line breaks.
- Exported values correspond to the source history and are not changed by display formatting unless specified.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R23-T1	History: 2+3 → 5 and 7×6 → 42; export CSV	Two data rows preserve expressions and results; proposed columns are timestamp, expression, result (confirm with team).
R23-T2	Export a record whose expression or note contains a comma	The CSV parses as one record with the comma-containing field intact.

UI / workflow notes: Offer text/CSV choice, a clear export action, and completion/failure feedback; avoid implying a download succeeded until confirmed.

Validation, limitations, and open decisions: Define included fields, timestamp format/time zone, character encoding, file naming, selection scope, and privacy expectations.

Dependencies and risks: Depends on history records (ID 3), possibly notes (ID 33), and file export support. Risk: ambiguous or incorrectly escaped data is hard to verify downstream.

Proposed definition of done: Export formats and fields are approved; representative and special-character files parse correctly; actual exported file evidence is attached.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 24 — Graph comparison

Proposed / hypothetical • Story points: 3 • Priority: 24 • Sprint: 2

Source description: An graph comparison view that lets multiple equations be displayed either separately or overlapping simultaneously.

What it should do

Display multiple selected equations in separate graph panes or together in one overlaid coordinate view so users can compare their shapes.

Acceptance criteria

- Each selected equation is identifiable and plotted in the selected comparison mode.
- Separate mode gives each graph a readable pane; overlay mode places graphs on shared axes.
- Changing comparison mode does not lose the selected equations.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R24-T1	Compare y=x and y=−x in separate mode	Two labeled panes show the rising and falling lines.
R24-T2	Switch the same two functions to overlay mode	Both lines appear on shared axes with distinguishable labels/styles.

UI / workflow notes: Show equation labels/legend and a clear separate/overlay control; maintain distinguishable line styling.

Validation, limitations, and open decisions: Define maximum graph count, shared vs independent axis scaling, color accessibility, and behavior for incompatible domains/ranges.

Dependencies and risks: Builds on graphing (ID 6), customization (ID 19), and UI settings (ID 22). Risk: overlapping plots can obscure each other or confuse axis scales.

Proposed definition of done: Comparison modes and scale rules are agreed; both functions remain identifiable in each mode; evidence includes separate and overlay screenshots.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 25 — Financial scenario comparison

Proposed / hypothetical • Story points: 3 • Priority: 25 • Sprint: 2

Source description: A scenario feature that allows users to save multiple financial calculations and compare results side by side.

What it should do

Save named financial input sets and results, then compare their outputs side by side using the same clearly labeled fields.

Acceptance criteria

- A scenario preserves its inputs, assumptions, and calculated result.
- At least two scenarios can be viewed together with matching fields aligned.
- Changing one scenario does not silently change another.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R25-T1	Save $1,000 at 5% annually for 2 years and $1,000 at 7% for 2 years, annual compounding	Scenario A = $1,102.50; B = $1,144.90; side-by-side labels show rates and terms.
R25-T2	Edit the rate in Scenario B after saving both	Scenario A retains its original 5% inputs and $1,102.50 result.

UI / workflow notes: Provide named scenario cards or columns with aligned input, assumption, and result labels; make save/compare actions explicit.

Validation, limitations, and open decisions: Define which finance calculations can be saved, maximum scenarios, duplicate names, edit/delete, and whether saved results recalculate when formulas/rates change.

Dependencies and risks: Depends on financial calculations (ID 9), persistent storage, and possibly favorites (ID 21). Risk: comparing results based on different assumptions without surfacing those assumptions.

Proposed definition of done: Scenario fields and update rules are agreed; save, compare, and isolation cases pass; captured evidence shows inputs and outputs together.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 26 — Programmer mode

Proposed / hypothetical • Story points: 2 • Priority: 26 • Sprint: 2

Source description: A programmer mode that allows users to convert numbers between decimal, binary, hex, and use bitwise logic.

What it should do

Convert integer values among decimal, binary, and hexadecimal representations and perform the approved bitwise operations.

Acceptance criteria

- Conversions preserve the same integer value across supported bases.
- Each supported bitwise operation returns the correct result under the selected width/signedness policy.
- Invalid digits for a base are rejected clearly.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R26-T1	Convert decimal 10 to binary and hexadecimal	Binary 1010; hexadecimal A.
R26-T2	Binary 1010 AND binary 1100	Binary 1000 (decimal 8) under ordinary bitwise AND.

UI / workflow notes: Make the active base visible beside input and output; label bitwise operation and operand base.

Validation, limitations, and open decisions: Define integer width, signedness, negative-number representation, leading zeros, overflow, and supported bitwise operators.

Dependencies and risks: Uses integer parsing/formatting and mode switcher (IDs 2 and 7). Risk: outputs differ substantially between signed/unsigned or fixed/unbounded widths.

Proposed definition of done: Base and bit-width rules are approved; conversions, bitwise operations, invalid digits, and boundary values pass; actual representations are captured.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 27 — Matrix operations

Proposed / hypothetical • Story points: 2 • Priority: 27 • Sprint: 2

Source description: A matrix tool that allows users to add, multiply, and find the inverse.

What it should do

Accept valid matrices and perform addition, multiplication, and inverse operations where the required dimensions and invertibility conditions hold.

Acceptance criteria

- Addition requires matching dimensions and adds corresponding elements.
- Multiplication requires compatible dimensions and uses row-by-column products.
- Inverse is returned only for a square nonsingular matrix.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R27-T1	A=[[1,2],[3,4]], B=[[5,6],[7,8]]; A+B	[[6,8],[10,12]].
R27-T2	A×B for the same matrices	[[19,22],[43,50]].
R27-T3	Inverse of A=[[1,2],[3,4]]	[[-2,1],[3/2,−1/2]]; $A \times A^{-1}$ = identity.
R27-T4	Inverse of [[1,2],[2,4]]	Singular-matrix error; no inverse.

UI / workflow notes: Provide a clear matrix-entry grid, show dimensions, and format output rows/columns consistently.

Validation, limitations, and open decisions: Reject ragged matrices and incompatible dimensions. Define maximum size, numeric precision, and how near-singular matrices are handled.

Dependencies and risks: Uses numeric engine (ID 2), matrix input UI, and display precision (ID 14). Risk: rounding near singularity may affect inverse decisions.

Proposed definition of done: Dimension and singularity rules are documented; operation, identity-check, and invalid-dimension cases pass; actual matrices and precision are recorded.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 28 — Calculus tools

Proposed / hypothetical • Story points: 1 • Priority: 28 • Sprint: 2

Source description: A calculus tool that allows users to find the derivatives and definite integrals.

What it should do

Compute derivatives and definite integrals for the team-approved class of functions and display the result with its variable and bounds.

Acceptance criteria

- A supported derivative result matches the approved symbolic or numeric convention.
- A definite integral uses the entered bounds and reports units/precision as applicable.
- Unsupported functions or invalid bounds produce an explanatory result.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R28-T1	Differentiate $f(x)=x^2$	2x under the proposed exact symbolic behavior.
R28-T2	Evaluate $\int_0^1 x^2\,dx$	1/3 under the proposed exact symbolic behavior; a numeric implementation may use an agreed tolerance.

UI / workflow notes: Separate derivative input from definite-integral input; show variable and integration bounds clearly.

Validation, limitations, and open decisions: Define supported functions, symbolic vs numeric behavior, numerical tolerance, singularity handling, and integration domain. The sample outputs assume exact elementary symbolic cases.

Dependencies and risks: Needs parser/function library (ID 1), numeric engine (ID 2), and possibly a symbolic algebra or numerical method. Risk: numerical approximations need a visible tolerance.

Proposed definition of done: Supported function class and accuracy policy are approved; known derivative/integral and invalid-domain cases pass; evidence includes method and actual result.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 29 — Complex-number mode

Proposed / hypothetical • Story points: 1 • Priority: 29 • Sprint: 2

Source description: Complex number mode that allows users to do standard math using real and imaginary numbers.

What it should do

Evaluate the team-approved standard arithmetic operations for values with real and imaginary components and display results in a consistent complex form.

Acceptance criteria

- Addition and multiplication combine real and imaginary components correctly.
- The display distinguishes the imaginary unit from a variable or ordinary text.
- Division by zero complex value is reported safely if division is included.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R29-T1	(1+2i) + (3−i)	4 + i.
R29-T2	(1+2i) × (3−i)	5 + 5i.

UI / workflow notes: Make complex input syntax and result format visible; preserve signs and omit zero components only according to a documented format.

Validation, limitations, and open decisions: Define supported operations, syntax (i or j), parsing precedence, display precision, and whether roots/logarithms are in scope.

Dependencies and risks: Needs parser and numeric engine support (IDs 1–2), with separate complex arithmetic semantics. Risk: syntax conflicts with variable names or scientific notation.

Proposed definition of done: Complex syntax and operation set are approved; arithmetic and error cases pass; examples show both components and actual outputs.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 30 — Find graph intercepts

Proposed / hypothetical • Story points: 2 • Priority: 30 • Sprint: Unassigned

Source description: Find intercept points on a graph for user

What it should do

Find and mark the x- and y-intercepts of a plotted supported function, and report coordinates in a readable form.

Acceptance criteria

- The y-intercept is evaluated at x=0 when defined.
- x-intercepts satisfy y=0 within the approved symbolic/numeric tolerance.
- Intercept markers/coordinates correspond to the displayed equation.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R30-T1	Find intercepts for $y=x^2-4$	x-intercepts $(-2,0)$ and $(2,0)$; y-intercept $(0,-4)$.
R30-T2	Find real x-intercepts for $y=x^2+1$	No real x-intercepts.

UI / workflow notes: Offer an intercept action in the graph view and label each marker/coordinate; distinguish “none found” from “not supported.”

Validation, limitations, and open decisions: Define root-finding tolerance, search range, repeated roots, complex roots, and behavior when a function is undefined at x=0.

Dependencies and risks: Builds on graphing (ID 6), expression parser (ID 1), and graph range settings (ID 19). Risk: numerical search can miss roots outside its interval.

Proposed definition of done: Search range/tolerance are documented; known roots and no-real-root cases pass; plotted markers agree with reported coordinates; evidence is attached.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 31 — Constants library

Proposed / hypothetical • Story points: 2 • Priority: 31 • Sprint: Unassigned

Source description: Constants Library (Speed of light, Gravity, etc.)

What it should do

Provide a searchable/selectable library of named scientific constants with value, unit, and a short identifying label.

Acceptance criteria

- A user can find a constant by its approved name or symbol.
- Selecting a constant inserts or displays its value with units.
- The stored value and precision follow a documented source/reference policy.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R31-T1	Look up c, speed of light in vacuum	299,792,458 m/s under the proposed exact SI value.
R31-T2	Look up standard gravitational acceleration g	9.80665 m/s² if the team adopts this conventional standard value; show units and source.

UI / workflow notes: Show name, symbol, value, and unit; allow search/filter and a clear insertion action.

Validation, limitations, and open decisions: Approve the constant list, aliases, precision, units, and authoritative source. g can vary by context; the test value is a proposed conventional standard, not a universal local measurement.

Dependencies and risks: Uses function/constant catalog and parser (IDs 1 and 8), plus unit display (ID 4). Risk: hidden precision or ambiguous constants may yield misleading calculations.

Proposed definition of done: Approved constants and sources are reviewed; lookup/insertion and unit cases pass; each value's provenance and precision appear in evidence.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 32 — Save graphs as PNG

Proposed / hypothetical • Story points: 1 • Priority: 32 • Sprint: Unassigned

Source description: Allow users to save graphs as a PNG

What it should do

Export the current graph view to a PNG image that preserves the plotted content and approved labels.

Acceptance criteria

- A PNG file is created after an explicit save action and can be reopened as an image.
- The exported graph contains the selected plot, axes, and visible labels.
- Export failure is reported without claiming a file was saved.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R32-T1	Plot y=x and save as PNG	A readable PNG opens and shows the line, axes, and graph label.
R32-T2	Attempt save to an unwritable/cancelled destination	Show a failure or cancellation state; do not report a successful export.

UI / workflow notes: Provide a save/export control and an explicit completion/failure message; let the user choose the destination.

Validation, limitations, and open decisions: Define pixel dimensions, background/transparency, filename, scale, and which overlays/controls are included.

Dependencies and risks: Depends on graph rendering (ID 6), custom graph state (ID 19), and comparison view (ID 24). Risk: saved image may not match visible graph if rendering uses stale settings.

Proposed definition of done: PNG scope and dimensions are agreed; exported file opens and matches the visible state; failure handling is checked; actual image evidence is attached.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 33 — Notes on saved calculations

Proposed / hypothetical • Story points: 1 • Priority: 33 • Sprint: Unassigned

Source description: Allow for short notes to be added to saved calculations

What it should do

Attach a short user-authored note to a saved calculation and show that note with the correct saved item when recalled.

Acceptance criteria

- A user can add and save a note on a saved calculation.
- The note remains associated with that calculation after recall.
- Editing one note does not change another saved calculation's note.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R33-T1	Add note "check units" to saved 2 + 3; reopen the saved item	The note "check units" appears on 2 + 3.
R33-T2	Save a second calculation without a note	The second item has no inherited "check units" note.

UI / workflow notes: Place a concise notes field on the saved-calculation detail and show the note in the recalled item.

Validation, limitations, and open decisions: Define maximum length, line breaks, supported characters, editing/deletion, and whether notes are included in exports.

Dependencies and risks: Requires saved calculations/favorites (ID 21) or equivalent persistence. Risk: notes may attach to a history row that later changes or is deleted.

Proposed definition of done: Note length and lifecycle rules are agreed; association, persistence, and isolation cases pass; evidence shows the correct note on the correct item.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 34 — Light and dark mode

Proposed / hypothetical • Story points: 1 • Priority: 34 • Sprint: Unassigned

Source description: Light/Dark Mode

What it should do

Let users switch the app between light and dark visual themes while preserving readable controls, text, graphs, and focus indicators.

Acceptance criteria

- Selecting Light applies the approved light palette.
- Selecting Dark applies the approved dark palette.
- Theme changes do not alter calculation input, results, or saved data.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R34-T1	Switch Light → Dark while result 2 + 3 = 5 is visible	Dark theme is applied; expression and result remain unchanged.
R34-T2	Switch Dark → Light	Light theme is applied and controls/text remain readable.

UI / workflow notes: Provide a clearly labeled theme control and consistent colors across calculator, dialogs, and graph views.

Validation, limitations, and open decisions: Define whether the theme follows system settings, persists across restart, and how customized colors (ID 22) interact. Confirm contrast and focus visibility.

Dependencies and risks: Uses shared design tokens from UI/customization (IDs 5 and 22). Risk: graph and dialog surfaces may retain hard-coded colors.

Proposed definition of done: Theme behavior and persistence are agreed; both palettes are checked across key views; calculation state is unchanged; screenshots document each theme.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 35 — Region-dependent number formatting

Proposed / hypothetical • Story points: 1 • Priority: 35 • Sprint: Unassigned

Source description: A region-dependent system that correctly displays decimals and commas.

What it should do

Format numeric inputs and results according to an approved region or locale, distinguishing decimal symbols from grouping separators.

Acceptance criteria

- Selecting locale displays the correct decimal and grouping symbols.
- Input parsing distinguishes decimal symbols from grouping separators without changing numeric value.
- Switching locales changes presentation without changing stored results.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R35-T1	Select en-US; display 1234.5	1,234.5 is displayed
R35-T2	Select de-DE; display 1234.5	1.234,5 is displayed.
R35-T3	Change locale after calculating 1234.5	Stored result stays 1234.5; only formatting changes.

UI / workflow notes: Offer an identifiable locale selection; show the active locale near relevant number inputs.

Validation, limitations, and open decisions: Decide supported locales, how pasted numbers are parsed, and whether locale settings persist. Ambiguous input should be rejected or explained.

Dependencies and risks: Depends on expression input (ID 1), numeric engine (ID 2), and display formatting (ID 14). Risk: comma decimal notation conflicts with function-argument separators.

Proposed definition of done: Supported locales and parsing policy are documented; locale and preservation tests pass; screenshots show both formats.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 36 — Right-click shortcut menu

Proposed / hypothetical • Story points: 1 • Priority: 36 • Sprint: Unassigned

Source description: A right click selection UI menu containing shortcut buttons such as copy and paste.

What it should do

Provide a context menu on relevant editable fields with working copy and paste actions.

Acceptance criteria

- Right-click on a supported input opens the menu in the appropriate position.
- Copy copies selected text or the documented field value.
- Paste inserts clipboard text at the cursor or replaces selected text, subject to browser permissions.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R36-T1	Right-click in an expression field	Context menu appears with Copy and Paste.
R36-T2	Select 2 + 3 and choose Copy	Clipboard contains the selected expression.
R36-T3	Choose Paste in a writable field	Permitted clipboard text is inserted at the cursor

UI / workflow notes: Use a compact context menu with clear labels, dismissal on Escape/outside click, and keyboard access where possible.

Validation, limitations, and open decisions: Define supported fields, selection behavior, clipboard permissions, and fallback if browser APIs deny clipboard access.

Dependencies and risks: Uses shared UI (ID 5) and expression editing (ID 12). Risk: native browser context menus and clipboard permissions differ by platform.

Proposed definition of done: Context menu works on approved fields; copy/paste and dismissal tests pass in supported browsers; evidence is attached.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 37 — Thousands separator toggle

Proposed / hypothetical • Story points: 1 • Priority: 37 • Sprint: Unassigned

Source description: A button to toggle the display of thousands separators in large numbers.

What it should do

Let users turn digit grouping on and off for displayed calculator values without changing the underlying number.

Acceptance criteria

- Toggle ON adds grouping separators for sufficiently large numbers under the active locale.
- Toggle OFF removes grouping separators from displayed values.
- Changing the toggle does not change numeric results, saved history values, or calculation accuracy.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R37-T1	Display 1234567.89; toggle ON in en-US	1,234,567.89 appears.
R37-T2	Toggle OFF for the same value	1234567.89 appears.
R37-T3	Toggle grouping after saving a calculation	Stored numeric result remains unchanged.

UI / workflow notes: Place a clearly labeled grouping toggle near display preferences; keep its current state visible.

Validation, limitations, and open decisions: Specify threshold for grouping, locale interaction (ID 35), negative numbers, decimals, and whether the preference persists.

Dependencies and risks: Depends on formatting controls (ID 14), locale formatting (ID 35), and shared UI (ID 5). Risk: formatting may accidentally be used as calculation input.

Proposed definition of done: Grouping behavior is documented; ON/OFF and state-preservation tests pass; results are verified against locale conventions.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 38 — Degree/radian angle toggle

Proposed / hypothetical • Story points: 1 • Priority: 38 • Sprint: Unassigned

Source description: An angle unit toggle button that allows the user to switch between degrees and radians.

What it should do

Provide an explicit angle-mode control that determines how supported trigonometric functions interpret angles.

Acceptance criteria

- The active DEG or RAD mode is clearly visible
- Trig calculations use the selected mode consistently
- Changing angle mode does not erase the current expression or saved history.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R38-T1	Select DEG; calculate sin(90)	Result is 1 within approved tolerance.
R38-T2	Select RAD; calculate sin(pi/2)	Result is 1 within approved tolerance.
R38-T3	Switch DEG to RAD with draft input present	Draft remains intact; active mode changes.

UI / workflow notes: Provide a visible DEG/RAD control in scientific mode and identify the active setting near results.

Validation, limitations, and open decisions: Define default mode, persistence, inverse trig output units, and treatment of saved calculations made in a different mode

. Dependencies and risks: Depends on scientific parser/functions (ID 1), calculation engine (ID 2), and mode switcher (ID 7). Risk: silently mixing angle conventions produces incorrect results.

Proposed definition of done: Angle-mode policy is approved; DEG/RAD calculation and state-preservation tests pass; active mode is evident in screenshots.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]

Requirement 39 — Scientific notation toggle

Proposed / hypothetical • Story points: 1 • Priority: 39 • Sprint: Unassigned

Source description: A scientific notation button to toggle the display of very large or small numbers as exponents.

What it should do

Allow users to switch numeric results between standard decimal display and scientific notation, subject to defined display limits.

Acceptance criteria

- Scientific mode displays supported values in coefficient × 10^exponent or equivalent E notation.
- Turning the toggle off restores ordinary decimal display where supported.
- Toggling notation changes only the display, not the stored numeric value.

Example test cases — proposed, not run

Test ID	Input / action	Expected output
R39-T1	Display 123000000; turn scientific notation ON	1.23 × 10^8 (or 1.23e8) appears.
R39-T2	Display 0.0000456; turn scientific notation ON	4.56 × 10^-5 (or 4.56e-5) appears.
R39-T3	Turn scientific notation OFF for 123000000	123000000 appears, subject to grouping preference.

UI / workflow notes: Show a clearly labeled SCI/EXP toggle and a visible active state; retain copyable numeric results.

Validation, limitations, and open decisions: Define notation threshold, significant digits, rounding, exponent limits, and interaction with precision (ID 14) and grouping (ID 37).

Dependencies and risks: Depends on numeric engine (ID 2) and display formatting (ID 14). Risk: a rounded formatted string may be reused as an internal result.

Proposed definition of done: Display rules and thresholds are documented; large/small number tests pass; stored values are unaffected; evidence is attached.

Evidence to add after verification: [Attach real test output, screenshot, demo link, build/commit, date, and observed result. Leave unfilled until evidence exists.]